

Team 2 - Billy's Amazing Team

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Following a series of group and client meetings, we categorised requirements into four key sections: User Requirements, System Requirements, Functional Requirements, and Non-Functional Requirements. This document outlines each category's essential requirements to ensure clarity and focus on delivering an engaging, user-friendly experience.

User Requirements

Based on three primary objectives from the Product Brief, we identified 12 critical user requirements aimed at creating a campus that is functional, intuitive, and enjoyable. To accommodate a diverse user base, accessibility options were also suggested, allowing players to adjust elements of the game, further aligning with our primary objectives.

Functional Requirements

The main objective of the game is to maximise student satisfaction through strategic campus design and building placement. 18 functional requirements were developed to support this, focusing on elements directly tied to user needs.

Non-Functional Requirements

To support an enjoyable experience, we established system performance standards. These allow us to facilitate future maintenance, making the system accessible to developers for edits and updates.

User Requirements

ID	Description	Priority
UR_UX	The user experience is pleasant	Shall
UR_BUILDINGS	The user can place buildings	Shall
UR_UI	The user interface is clear and pleasant to look at	Shall
UR_DISPLAY_TIMER	The game shall display a visible timer to the user to keep track of time	Shall
UR_BUILDING_COUNTER	The game shall display a counter showing the number of each type of building placed	Shall
UR_PAUSE	The user can pause the game at anytime	Shall
UR_UPGRADE	The user can upgrade buildings	Maybe
UR_SATISFACTION_METER	The system shall display the student satisfaction, which then has its change.	Shall
UR_ACCESSABILITY_SETTINGS	The user can choose from some accessibility settings	Should
UR_EVENTS	The user will get a pop up explaining the current event	Shall
UR_KEYBINDS	The user can use keybinds to quickly perform an action	Maybe
UR_SCOREBOARD	The user can see the top 5 scores achieved at the end of the game.	Shall
UR_ACHIEVEMENTS	The user can complete a number of achievements by fulfilling certain criteria in the game.	Shall

Functional Requirements

ID	Description	User Requirements
FR_SYS_REPONSE	The system responds to user inputs clearly	UR_UI

FR_BUILDING_PLACEMENT	The system allow the user to place buildings in buildable areas only	UR_BUILDINGS
FR_BUILDING_SATISFACTION_GAIN	After a building is built it will increase the satisfaction for a period of time	UR_BUILDINGS
FR_CONSTRUCTION_SATISFACTION_LOSS	Building/upgrading a building during term time will result in satisfaction loss for a period of time	UR_BUILDINGS UR_UPGRADE
FR_TIMER_DISPLAY	The game shall implement a timer counting down from 5 minutes in real time, allowing the user to monitor the time elapsed	UR_DISPLAY_TIMER
FR_BUILDING_COUNTER	The game shall include a counter that updates whenever a building is placed, showing the number of each type	UR_BUILDING_COUNTER
FR_SATISFACTION_BREAK_DOWN	The system shall display what is affecting the student satisfaction, in the form of student thoughts displayed as text on the screen	UR_SATISFACTION_METER
FR_BUILDING_OVERLAP	The system shall not allow a building to be placed if it will overlap with another building	UR_BUILDINGS
FR_BUILDING_CONSTRUCTION	Buildings take time to be constructed before they can have a positive satisfaction gain	UR_BUILDINGS
FR_SATISFACTION_TRACK	The system shall display where the satisfaction will end up over time.	UR_SATISFACTION_METER
FR_COLOUR_BLIND_MODE	The system shall allow the user to select if they want a colour blind mode on.	UR_ACCESSABILITY_SETTINGS
FR_MOUSE_SENSITIVITY_SLIDER	The system shall allow the user to choose a mouse sensitivity	UR_ACCESSABILITY_SETTINGS
FR_RANDOM_EVENT	When there is not an event cooldown, a random event can happen from a set list.	UR_EVENT
FR_SET_EVENT	If the events are not on cooldown, then at a certain time a set event will happen	UR_EVENT

FR_SATISFACTION_PAUSE	When the game is paused, the satisfaction will not change.	UR_PAUSE
FR_TIMER_PAUSE	When the game is paused, the timer will not change.	UR_PAUSE
FR_MULTIPLE_BUILDINGS	When there are multiple buildings being built/upgraded at once, there is an increasing multiplier on the satisfaction modifier from the building's construction during term time.	UR_BUILDINGS
FR_UPGRADE_SATISFACTION_GAIN	When a building's upgrade has finished construction, satisfaction will increase for a period of time.	UR_UPGRADE
FR_CALCULATE_SCORE	Score must be calculated from satisfaction during the game.	UR_SCOREBOARD
FR_DISPLAY_SCORES	The system will display the current score at all times as well as displaying the final score when the game ends compared to the top 5 high scores recorded locally.	UR_SCOREBOARD
FR_STORE_SCORE	Any score which exceeds any of the previous top 5 scores must be locally recorded in the correct place in the high score list.	UR_SCOREBOARD
FR_CHECK_ACHIEVEMENTS	If the user reaches or exceeds the requirements for any of the game's achievements it should be displayed.	UR_ACHIEVEMENTS

Non-Functional Requirements

ID	Description	User Requirements	Constraints
NFR_SYS_RESPONSE_SPEED	The system responds to user inputs quickly.	UR_UX	In <0.1s from time of input 90% of the time
NFR_COUNTER_VISIBILITY	The building counter shall update in real time whenever a building is placed.	UR_BUILDING_COUNTER	Updates within 1 second of placement
NFR_TIMER_VISIBILITY	The timer shall be visible and readable at all times during gameplay	UR_DISPLAY_TIME	Timer visibility is ensured throughout play
NFR_SATISFACTION_MULT	Building satisfaction gain/loss shall be multiplied	UR_BUILDINGS	

NFR_SATISFACTION_METER_VISIBILITY	The satisfaction meter shall be visible and readable at all times	UR_SATISFACTION_METRE	Satisfaction meter is visible 99% of the time
NFR_READABILITY	All user-facing messages shall be in plain- english and without technical jargon	UR_UX	This should be the case 95% of the time
NFR_EVENT_COOLDOWN	There shall be period where an event can not happen after an event has happened	UR_EVENTS	The cooldown shall be 1 minute.
NFR_MULTIPLE_CONSTRUCTION	There shall be no limit on the number of buildings the player can build/upgrade at once	UR_BUILDINGS	